

DGX Spark paired governor analysis

A controlled comparison of four GPU governors under CPU powersave (858 MHz) and CPU performance (3354 MHz).

Primary metric: median of per-run GPU EDP = `avg_gpu_time_sec` squared times `power_draw_w_avg`. Application latency is `avg_launch_time_sec` and is only the `<=5%` constraint.

- 2,400 runs | 240 ten-run cells | 30 applications | 110 valid paired CPU-mode EDP comparisons
- Median performance-vs-powersave effect: application latency -2.76%; GPU EDP -0.17%.
- Missing GPU time is censored. Zero GPU time is instrumentation-limited. Neither is zero energy.

Method and decision rule

For each run, calculate GPU EDP from GPU time and average board power. For each 10-run cell, take the median of those per-run products. Compare each non-default GPU governor with default inside the same CPU mode.

A robust recommendation must be EDP-valid and have application latency no more than 5% above default in both CPU modes. The selected candidate minimizes the geometric mean EDP ratio across CPU modes.

- avg_launch_time_sec is never an EDP input.
- CPU EDP is omitted because CPU package power is unavailable.
- All percentage rankings exclude censored and zero-GPU-time cells.

Experimental matrix and measurement status

Each tile is one 10-run cell; gray is censored, amber is zero GPU timer

	Powersave Default	Powersave Ondemand	Powersave Powersave	Powersave Performance	Performance Default	Performance Ondemand	Performance Powersave	Performance Performance
babelstream								
backprop								
bfs								
bh								
cfd								
fft								
gaussian								
gemmEx								
heartwall								
hotspot								
hybridsort								
ising								
kmeans								
lavaMD								
leukocyte								
lud								
lulesh								
memcpy								
mixbench								
myocyte								
nbody								
nn								
nw								
particlefilter								
pathfinder								
srad								
stencil1d								
stencil3d								
streamcluster								
triad								

VALID

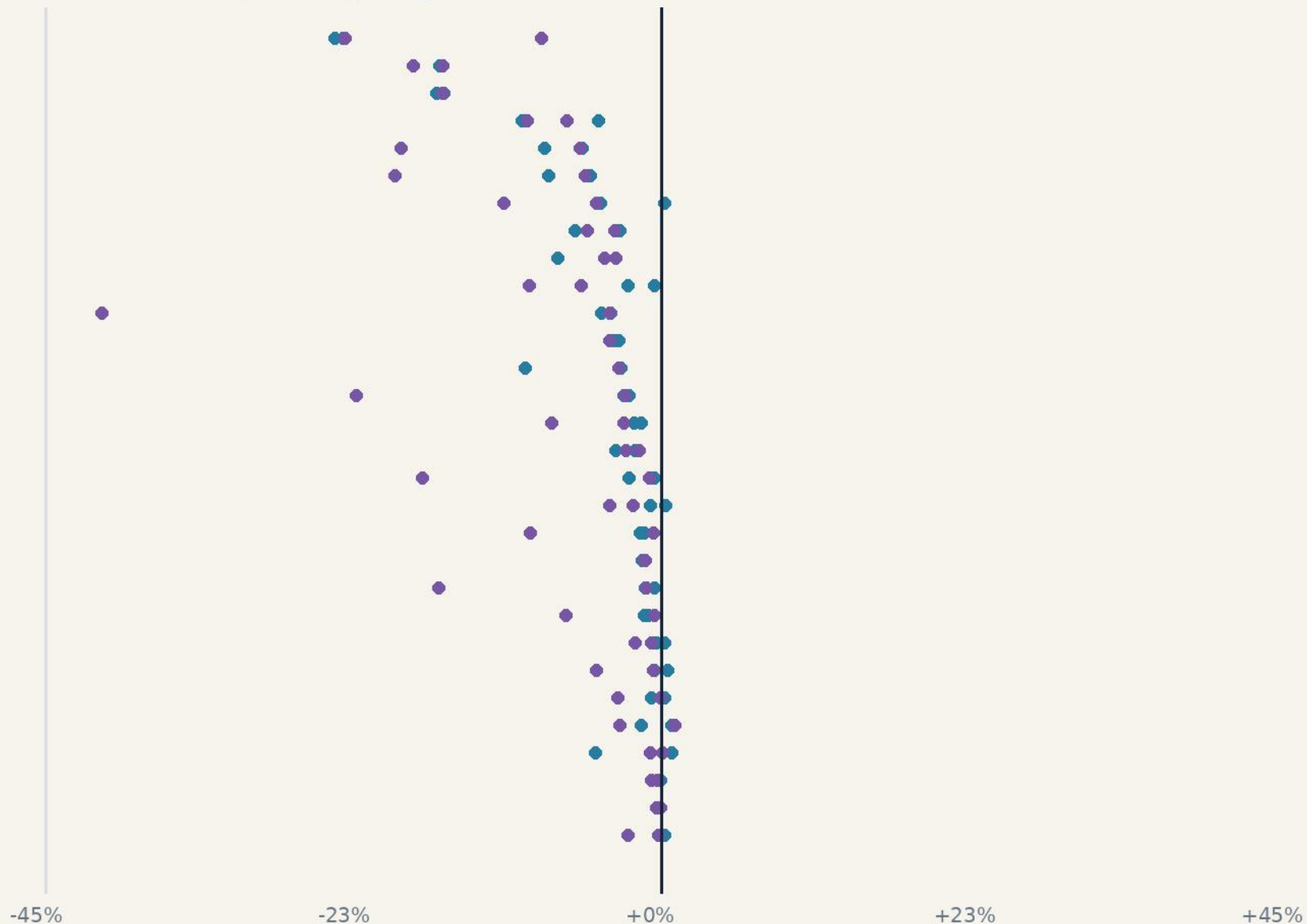
CENSORED FAILURE

ZERO-TIME / INSTRUMENTATION LIMITED

CPU governor impact on application latency

Performance vs powersave; each dot is one GPU-governor pairing

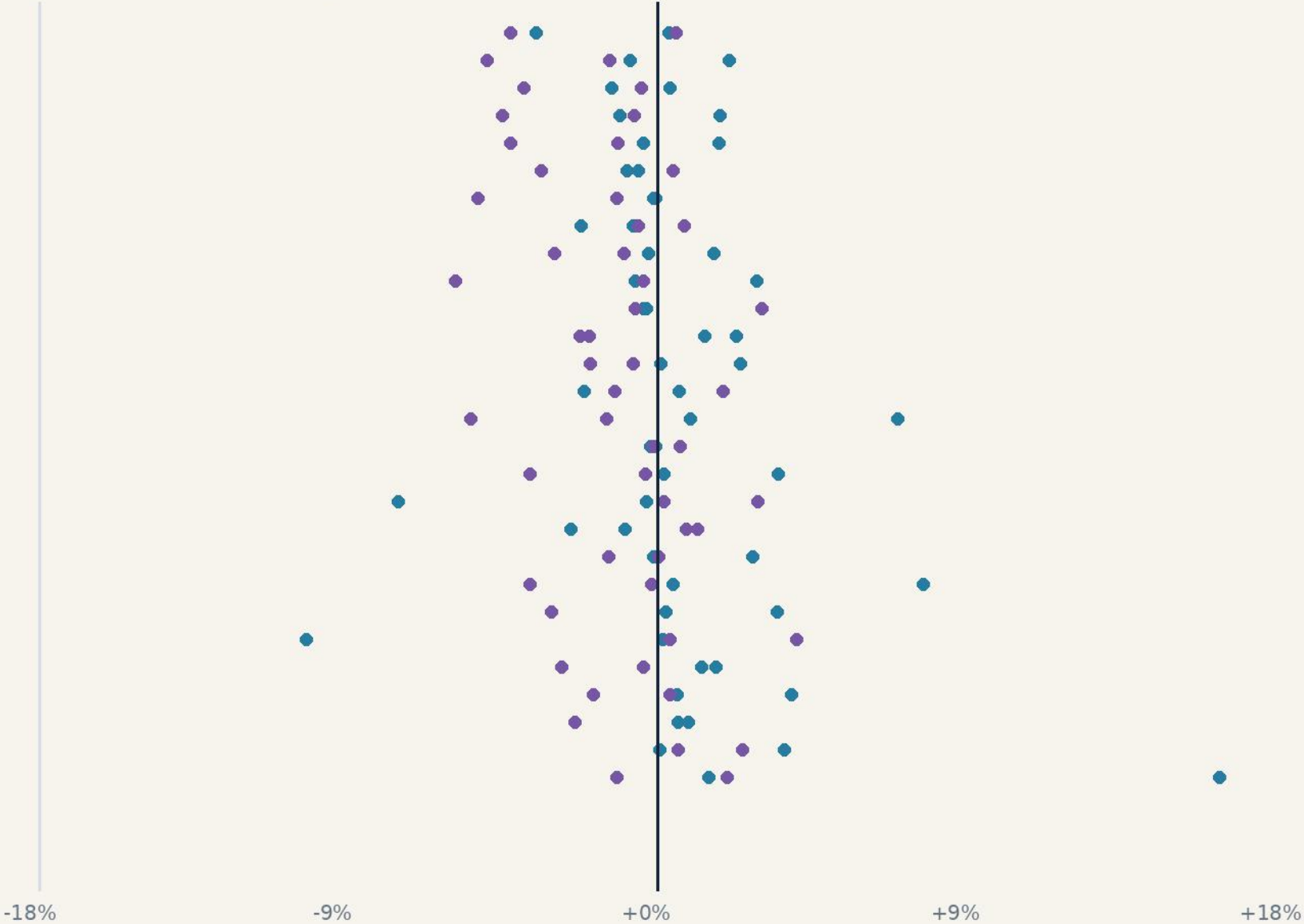
babelstream
lavaMD
gemmEx
nw
pathfinder
srad
stencil3d
hybridsort
lulesh
gaussian
nbody
mixbench
myocyte
stencil1d
kmeans
bfs
fft
cfd
particlefilter
nn
bh
ising
heartwall
hotspot
lud
memcpy
streamcluster
triad
leukocyte
backprop



CPU governor impact on GPU EDP

Performance vs powersave; valid positive GPU timers only

- fft
- backprop
- stencil1d
- stencil3d
- heartwall
- bh
- pathfinder
- lavaMD
- kmeans
- bfs
- lud
- hybridsort
- leukocyte
- srad
- nw
- nbody
- gaussian
- hotspot
- babelstream
- nn
- triad
- mixbench
- cfid
- lulesh
- ising
- myocyte
- particlefilter
- streamcluster



GPU governor EDP change vs default

Median per-run GPU EDP; lower is better. Zero-time and censored cells are not ranked.

Powersave (858 MHz)

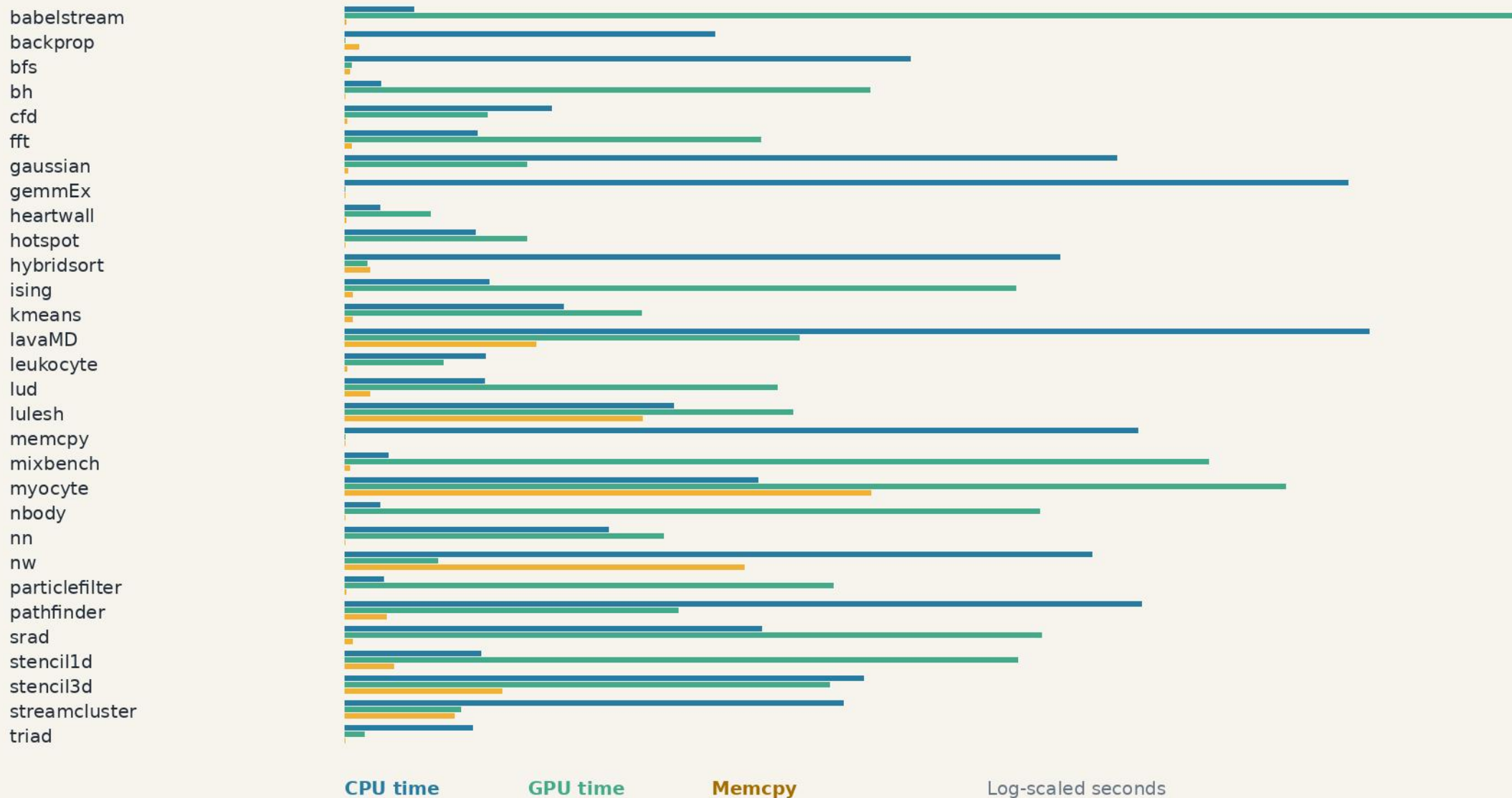
	Ondemand	Powersave	Performance
babelstream	+23.2%	-46.5%	+19.0%
backprop	-52.2%	-50.4%	+54.2%
bfs	+296.7%	+308.0%	+112.6%
bh	-0.4%	+1426.3%	+2.5%
cfcd	+2.5%	+753.1%	+11.9%
fft	+1.2%	+2101.5%	+3.4%
gaussian	+69.1%	+846.4%	+89.9%
gemmEx	NA	NA	NA
heartwall	-5.9%	+1730.7%	+4.0%
hotspot	+1.4%	+1048.5%	+4.4%
hybridsort	+657.8%	+654.0%	+72.0%
ising	+5.5%	+243.5%	+8.4%
kmeans	+19.8%	+1142.4%	+9.2%
lavaMD	-3.2%	+5155.1%	+59.2%
leukocyte	+132.1%	+3079.5%	+5.5%
lud	+16.7%	+12.7%	+20.0%
lulesh	+14.8%	+1313.0%	+15.8%
memcpy	NA	NA	NA
mixbench	-2.9%	NA	+4.0%
myocyte	+39.0%	NA	+1.4%
nbody	-0.2%	+805.2%	+0.1%
nn	-3.5%	+238.4%	+1.5%
nw	-41.1%	+804.1%	+53.2%
particlefilter	+14.5%	+1005.7%	+14.1%
pathfinder	+36.4%	+5030.4%	+66.4%
srad	+18.8%	+111.6%	+22.4%
stencil1d	+17.8%	+299.7%	+22.2%
stencil3d	+0.4%	+767.6%	+28.8%
streamcluster	-14.2%	-16.1%	+47.0%
triad	-24.6%	+18.4%	+12.0%

Performance (3354 MHz)

	Ondemand	Powersave	Performance
babelstream	+21.2%	-45.3%	+21.2%
backprop	-50.8%	-50.7%	+47.7%
bfs	+310.8%	+309.1%	+101.4%
bh	-0.7%	+1441.5%	-0.4%
cfcd	+14.3%	+888.5%	+25.1%
fft	-2.8%	+2105.7%	-1.3%
gaussian	+74.7%	+841.2%	+82.5%
gemmEx	NA	NA	NA
heartwall	-3.8%	+1716.9%	-0.0%
hotspot	+9.3%	+1178.0%	+13.1%
hybridsort	+664.7%	+629.2%	+65.9%
ising	+9.0%	+242.8%	+5.8%
kmeans	+22.1%	+1133.4%	+6.2%
lavaMD	-1.7%	+5316.6%	+61.9%
leukocyte	+137.5%	+3054.9%	+3.3%
lud	+16.9%	+16.6%	+19.7%
lulesh	+15.3%	+1289.7%	+11.2%
memcpy	NA	NA	NA
mixbench	+0.2%	NA	+0.6%
myocyte	+39.4%	NA	-1.6%
nbody	-0.1%	+812.9%	+0.2%
nn	-0.7%	+234.1%	+1.6%
nw	-37.6%	+782.3%	+43.5%
particlefilter	+10.5%	+992.6%	+10.6%
pathfinder	+36.4%	+4973.1%	+57.9%
srad	+22.2%	+120.3%	+23.5%
stencil1d	+15.9%	+296.3%	+17.0%
stencil3d	-2.5%	+746.3%	+20.8%
streamcluster	-25.1%	-26.4%	+24.9%
triad	-19.1%	+17.7%	+7.4%

Time-component and bottleneck view

Default governor medians; components are diagnostic and explicitly nonadditive



Valid GPU EDP wins under the latency guardrail

Count of applications with lower EDP and <=5% application-latency change vs default

Powersave (858 MHz)

On demand



6 wins | 14 non-wins | 10 unranked

Powersave



1 wins | 2 non-wins | 27 unranked

Performance



0 wins | 28 non-wins | 2 unranked

Performance (3354 MHz)

On demand



5 wins | 13 non-wins | 12 unranked

Powersave



1 wins | 2 non-wins | 27 unranked

Performance



4 wins | 24 non-wins | 2 unranked

Policy recommendation and cross-CPU validation

Recommended governor requires valid GPU EDP and <=5% latency in both CPU modes

babelstream	Default		+0.0%	worst latency +0.0%
backprop	Ondemand		-51.5%	worst latency +2.1%
bfs	Default		+0.0%	worst latency +0.0%
bh	Default		+0.0%	worst latency +0.0%
cfd	Default		+0.0%	worst latency +0.0%
fft	Default		+0.0%	worst latency +0.0%
gaussian	Default		+0.0%	worst latency +0.0%
gemmEx	Default	UNRANKABLE		worst latency +0.0%
heartwall	Default		+0.0%	worst latency +0.0%
hotspot	Default		+0.0%	worst latency +0.0%
hybridsort	Default		+0.0%	worst latency +0.0%
ising	Default		+0.0%	worst latency +0.0%
kmeans	Default		+0.0%	worst latency +0.0%
lavaMD	Ondemand		-2.5%	worst latency +0.0%
leukocyte	Default		+0.0%	worst latency +0.0%
lud	Default		+0.0%	worst latency +0.0%
lulesh	Default		+0.0%	worst latency +0.0%
memcpy	Default	UNRANKABLE		worst latency +0.0%
mixbench	Default		+0.0%	worst latency +0.0%
myocyte	Default		+0.0%	worst latency +0.0%
nbody	Default		+0.0%	worst latency +0.0%
nn	Default		+0.0%	worst latency +0.0%
nw	Default		+0.0%	worst latency +0.0%
particlefilter	Default		+0.0%	worst latency +0.0%
pathfinder	Default		+0.0%	worst latency +0.0%
srad	Default		+0.0%	worst latency +0.0%
stencil1d	Default		+0.0%	worst latency +0.0%
stencil3d	Default		+0.0%	worst latency +0.0%
streamcluster	Default		+0.0%	worst latency +0.0%
triad	Default		+0.0%	worst latency +0.0%

Findings and deployment

CPU performance changed median application latency by -2.76% and valid-pair GPU EDP by -0.17% versus CPU powersave. These are separate responses: CPU frequency can alter host-side latency without a proportional GPU EDP response.

2 applications have a robust lower-EDP recommendation. Full per-application values and validation are in recommendations.csv and FINDINGS.md.

- Deploy only PASS mappings; preserve default for unrankable timers.
- Validate positive GPU time, power samples, completed launches, utilization, clocks, thermals, problem size, and the +5% latency sentinel.
- Re-profile after driver, runtime, benchmark, thermal, or hardware changes.

Limitations

- CPU modes were measured on different dates; date and CPU mode are confounded.
- Component timers overlap and are nonadditive.
- CPU package power is unavailable, preventing CPU and whole-system EDP.
- Ten-run medians are descriptive and do not imply statistical significance.
- Conclusions apply to the measured large-input DGX Spark environment.